

Mid-Term Report: Sensor-Adaptive Agricultural Vision Foundation Models

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Abstract

This project investigates continual self-supervised pretraining of official Vision Transformer (ViT) and Yolo checkpoints on a diverse agricultural image corpus. Our aim is to improve transfer across crops, diseases, acquisition settings, and sensor modalities without substantially changing the pretrained backbone. Since the proposal, we have built a single training stack that supports ViT-S/B/L models initialized from ImageNet checkpoints; masked image modeling (MIM); DINO-style student-teacher training; and a learnable 1×1 spectral adapter for RGB and multiband inputs. The data pipeline can stream ordinary folders, ZIP and nested-ZIP archives, GeoTIFFs, NumPy arrays, and WebDataset shards. The validated catalog now contains 5,175,016 RGB images, up from 412,454 images at proposal time. We have verified end-to-end MIM and DINO training, five-band inputs, and a 474-epoch SparK-YOLO run whose best validation loss fell by 18.46%. These are encouraging engineering results, but they do not yet demonstrate better representations. Real multispectral data, duplicate control, source-balanced sampling, and held-out-domain comparisons are still being completed. The next phase will set ImageNet baselines and compare linear-probe, few-shot, and fine-tuning performance before larger runs begin.

1. Introduction

Agricultural imagery varies in ways that standard image benchmarks often do not capture. A crop can look different across growth stages, stress conditions, seasons, locations, management practices, cameras, and lighting. Disease benchmarks frequently contain centered leaf images, while practical systems must handle cluttered field scenes, unmanned aerial vehicle imagery, or multispectral rasters. Labels are also expensive to collect and are scattered across tasks such as species identification, disease diagnosis, weed recognition, detection, segmentation, counting, and temporal monitoring. A model that performs well on a random split of one dataset may therefore fail to transfer to a new farm or a different sensor.

We ask whether continual self-supervised pretraining can adapt an official ViT to agricultural visual structure while

retaining the benefits of generic pretraining. The study compares reconstruction-based MIM with view-invariant DINO-style self-distillation and will also test a sequential MIM→DINO schedule. A lightweight spectral adapter maps any number of aligned input bands to the three channels expected by an official RGB ViT. Keeping the backbone fixed makes the effect of sensor adaptation easier to measure.

At this midpoint, the project has a working research infrastructure and a much larger indexed corpus. The software supports both objectives, several official checkpoint families, archive-scale streaming, reproducible runs, and downstream task runners. The central scientific question, however, remains open. Passing tests and decreasing pretext losses show that the pipeline works; they do not show that it transfers better. For that reason, we separate engineering evidence from scientific results and defer performance claims until controlled downstream experiments are complete.

2. Background and Related Work

Vision Transformers. ViT treats an image as a sequence of patch tokens processed by a Transformer encoder. With sufficiently large-scale pretraining, this approach can match strong convolutional models [1]. We use official ViT-S, ViT-B, and ViT-L implementations rather than a custom transformer so that initialization and model scale remain controlled across experiments.

Masked image modeling. MAE hides a large fraction of image patches and trains an asymmetric encoder-decoder to reconstruct them, yielding transferable features [2]. Other masked-image methods similarly show that reconstruction or token prediction can provide a useful self-supervised signal [3]. SparK applies the same broad idea to convolutional encoders through a sparse hierarchy [4]. Agricultural lesions, leaf venation, edges, and canopy texture make this local reconstruction signal worth testing for domain adaptation.

Self-distillation. DINO trains a student to match the centered, sharpened outputs of an exponential-moving-average (EMA) teacher from different augmented views [5]. DINOv2 showed that careful curation and self-distillation can produce robust visual features without labels [6]. Because reconstruction and self-distillation emphasize different invariances, we evaluate MIM-only, DINO-only, and sequential training with the same data and backbone stack.

Table 1. Corpus status at proposal and mid-term. Counts describe indexed files, not unique biological observations.

Property	Proposal	Mid-term
Validated images	412,454	5,175,016
Local storage	35.86 GB	~1 TB
Primary modality	RGB	RGB
Real multispectral archive	No	No
Archive streaming	Audit only	ZIP/TAR/WDS

Agricultural transfer and spectral adaptation.

PlantVillage helped establish deep learning for image-based plant-disease recognition, but its controlled backgrounds also highlight the gap between benchmark and field conditions [7, 8]. Remote-sensing models such as SatMAE extend masked pretraining to temporal and multispectral satellite imagery [9]. Our approach instead keeps an official RGB backbone and learns a small sensor-specific projection. This gives us a direct comparison with RGB-only input and with patch-embedding expansion.

3. Approach

3.1. Data and Sampling

The proposal described 412,454 images from 13 downloaded datasets. The validated catalog now indexes 5,175,016 images visible to the loaders, stored across about 1 TB of ZIP and TAR archives. It includes plant species, disease, seedling, weed, and crop imagery, although nearly all samples are still RGB. Each record retains a stable sample key, source identifier, and portable archive-member path. For large sources, we create WebDataset shards of roughly 1 GB after RGB conversion, bounded resizing, and validation. The preprocessing workers handle one archive at a time, which avoids repeated archive opens and expensive seeks.

To keep a single large collection from dominating training, the planned sampler will choose a source first and a sample second. When source metadata permits, validation splits will be group-disjoint. Before the final experiments, we will freeze the manifest, record dataset versions and licenses, and remove exact and near-duplicate clusters before defining training and held-out-domain partitions.

3.2. Sensor-Adaptive Backbone

For an input $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$, a learnable 1×1 convolution projects the C sensor bands to the three channels expected by an official pretrained ViT:

$$\mathbf{z}_{h,w} = A\mathbf{x}_{h,w} + \mathbf{b}, \quad A \in \mathbb{R}^{3 \times C}. \quad (1)$$

For RGB, A is initialized to the 3×3 identity and $\mathbf{b} = 0$. For multiband input, available RGB-like channels initialize the corresponding outputs; the additional-band weights start

at zero but can be learned. We then normalize the adapted image using the statistics for the selected checkpoint and feed it to the unchanged ViT patch embedding. The adapter adds just $3C + 3$ parameters, separating sensor mixing from spatial feature learning.

3.3. Self-Supervised Objectives

In MIM, we mask a random set $\mathcal{M}$ containing 75% of the patch tokens. A linear reconstruction head predicts the adapted RGB patch $\mathbf{p}_i$ from the encoded token $\mathbf{h}_i$, and the loss is evaluated only at masked positions:

$$\mathcal{L}_{\text{MIM}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|g(\mathbf{h}_i) - \mathbf{p}_i\|_2^2. \quad (2)$$

For DINO-style training, the student receives global and local crops while the teacher receives paired global crops. With student temperature τ_s , teacher temperature τ_t , and running center $\mathbf{c}$, the cross-view loss is

$$\mathcal{L}_{\text{DINO}} = - \sum_k q_t^{(k)} \log q_s^{(k)}, \quad q_t = \text{softmax}((\mathbf{o}_t - \mathbf{c})/\tau_t). \quad (3)$$

The teacher parameters—adapter, backbone, and projection head—are updated by $\theta_t \leftarrow m\theta_t + (1 - m)\theta_s$. The implementation supports ImageNet, MAE, DINOv2, and DINOv3 initialization, as well as an explicit MIM→DINO handoff that begins a new optimization run from a compatible MIM checkpoint.

3.4. Training and Evaluation Protocol

Training uses BF16 mixed precision, gradient accumulation, optional gradient checkpointing, cosine learning-rate schedules, and multi-process data loading. Each run saves the resolved configuration, environment and Git metadata, metrics, figures, best and latest checkpoints, optimizer state, and random-number-generator state. WebDataset handles CPU streaming, while NVIDIA DALI is available on Linux for GPU JPEG decoding. Development began on a single RTX 4090; the next stage will use Auburn’s Slurm cluster for longer controlled runs.

The evaluation will compare the unadapted official checkpoint with RGB-only continual pretraining, MIM, DINO, sequential MIM→DINO, frozen and trainable adapters, and direct patch-embedding expansion. We will report accuracy and macro-F1 for classification; mAP, IoU/Dice, and MAE/RMSE will be used for detection, segmentation, and counting as appropriate. Linear probes, few-shot subsets, and full fine-tuning will each be repeated with at least three seeds. The main test sets will hold out entire sources, crops, geographic areas, seasons, or sensors, and shared splits will include means, standard deviations, and paired comparisons.

Table 2. Mid-term implementation validation. “Pass” indicates successful execution and expected artifacts, not a downstream accuracy result.

Check	Input	Result
MIM, official ImageNet ViT-S	5-band NPY	Pass
DINO, official ImageNet ViT-S	RGB PNG	Pass
DINO, official DINOv2 ViT-S	RGB PNG	Pass
MIM→DINO initialization	5-band NPY	Pass
Two-epoch checkpoint resume	RGB/5-band	Pass
Automated test suite	124 tests	124 pass
Shard builder smoke test	ZIP/plain	2 pass

4. Intermediate and Preliminary Results

4.1. Verified Engineering Results

Table 2 summarizes the completed checks. End-to-end runs produced checkpoints, metrics, training curves, MIM reconstructions, DINO augmented views, and student–teacher similarity matrices. Five-band float32 arrays successfully passed through the learnable adapter and an official ImageNet ViT-S. DINO also ran from an official DINOv2 ViT-S checkpoint using its patch-14 geometry, and a compatible MIM checkpoint initialized a new five-band DINO run. In two-epoch resume tests, the optimizer, scheduler, scaler, history, random state, and loader state were all restored.

4.2. Long-Run SparK–YOLO Pretraining

We also evaluated the convolutional branch with a 500-epoch SparK run using the WebDataset pipeline and a YOLO11-L backbone. We initialized from the official YOLO checkpoint, inserted 1×1 expansion bridges where the dense and sparse channel dimensions differed, and transferred 215 tensors to the sparse encoder. Each epoch contained 4,000 training batches. The learning rate warmed from 2×10^{-6} to 10^{-5} during the first five epochs, stayed near that value early in the run, and then decayed toward 10^{-6} on a cosine schedule.

When we analyzed the run, 474 epochs had completed and epoch 475 was 1,232 of 4,000 batches in progress. Validation loss decreased from 0.8071 at epoch 1 to 0.6581 at epoch 472, a relative reduction of 18.46%. Training loss at that point was 0.6524, for a gap of 0.0057. Table 3 lists selected checkpoints. Most of the improvement arrived in the first 100 epochs. After that, the curve improved more slowly but still reached new minima at epochs 168, 257, 297, 353, 411, 427, 452, and 472. Across the final 25 completed epochs, validation loss averaged 0.6610 with a standard deviation of 0.0014. This tight range suggests a stable late-stage plateau rather than divergence. The final observed validation loss (0.6615) was slightly above the minimum, so subse-

Table 3. Selected points from the SparK–YOLO pretraining log. Epoch 475 was still running and is therefore excluded.

Epoch	Train	Validation	LR
1	0.8250	0.8071	2×10^{-6}
100	0.6782	0.6808	9×10^{-6}
297	0.6601	0.6598	4×10^{-6}
472	0.6524	0.6581	1×10^{-6}
474	0.6431	0.6615	1×10^{-6}

quent experiments will use the saved best checkpoint rather than the terminal state.

The automated suite covers RGB, NPY, GeoTIFF, ZIP, and nested-ZIP loading; group-disjoint splits; NoData handling; padding for undersized images; official model selection; adapter initialization; MIM and DINO tensor behavior; EMA updates; paired-crop replay; strict configuration parsing; atomic checkpointing; and RoPE-based DINOv3 backbones. Recent shard-builder tests also check parallel archive grouping and valid TAR output. Together, these tests reduce the chance that a long pretraining run fails because of data-format, checkpoint, or model- geometry errors.

4.3. Interpretation and Remaining Gaps

The expanded corpus, working pipelines, and four-day SparK–YOLO run meet the mid-term engineering milestone: the planned models can now train reproducibly at archive scale. The long run gives useful evidence that sparse conversion, transferred YOLO weights, checkpoint selection, and WebDataset streaming remain numerically stable beyond a smoke test. It does not show that the representation improves a recognition task, since reconstruction loss can decrease without improving transfer.

The same official backbone also accepts RGB and five-band tensors without a replacement patch embedding, which supports the feasibility of the spectral interface. It does *not* tell us whether extra bands improve predictions. The five-band inputs used for verification are synthetic, and the catalog still lacks a verified real multispectral archive.

No model has yet outperformed its official ImageNet or DINO initialization on a held-out agricultural benchmark. Accordingly, we do not claim improved accuracy, robustness, or label efficiency at this stage. The main risks are duplicate leakage, imbalance from the 306k-image PlantNet subset and other large sources, RGB-only bias, and I/O saturation during multi-GPU training. We will address the first three through manifest freezing, duplicate clustering, source-balanced sampling, and external domain splits. Before requesting more GPUs, we will measure the remaining risk by comparing CPU WebDataset and DALI throughput.

The next milestone is a ViT-S experiment matrix on frozen data: official initialization, MIM, DINO, and se-

quential MIM→DINO. We will then run linear probes and low-label fine-tuning on source-disjoint disease and species benchmarks. Only approaches that improve transfer consistently across seeds will move on to ViT-B. In parallel, we will add a licensed real multispectral source and compare frozen and trainable adapters with direct patch-embedding expansion. This sequence keeps the conclusions tied to the evidence and to the data that are actually available.

References

- [1] A. Dosovitskiy et al. An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale. *ICLR*, 2021.
- [2] K. He et al. Masked Autoencoders Are Scalable Vision Learners. *CVPR*, 2022.
- [3] H. Bao et al. BEiT: BERT Pre-Training of Image Transformers. *ICLR*, 2022.
- [4] K. Tian, Y. Jiang, Q. Diao, C. Lin, L. Wang, and Z. Yuan. Designing BERT for Convolutional Networks: Sparse and Hierarchical Masked Modeling. *ICLR*, 2023.
- [5] M. Caron et al. Emerging Properties in Self-Supervised Vision Transformers. *ICCV*, 2021.
- [6] M. Oquab et al. DINOv2: Learning Robust Visual Features without Supervision. *TMLR*, 2024.
- [7] D. Hughes and M. Salathé. An Open Access Repository of Images on Plant Health to Enable the Development of Mobile Disease Diagnostics. *CoRR*, 2015.
- [8] S. Mohanty, D. Hughes, and M. Salathé. Using Deep Learning for Image-Based Plant Disease Detection. *Frontiers in Plant Science*, 2016.
- [9] Y. Cong et al. SatMAE: Pre-training Transformers for Temporal and Multi-Spectral Satellite Imagery. *NeurIPS*, 2022.